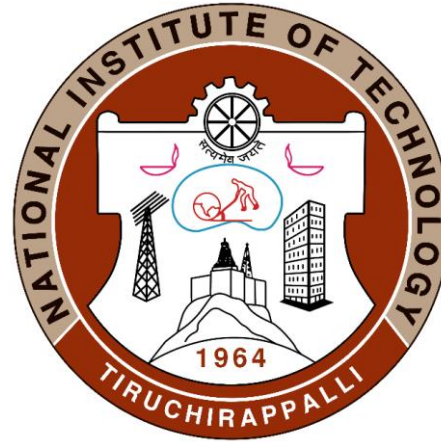


Polymers, Ceramics and Composites



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Lecture - 5

Conductive polymers

Conductive polymers

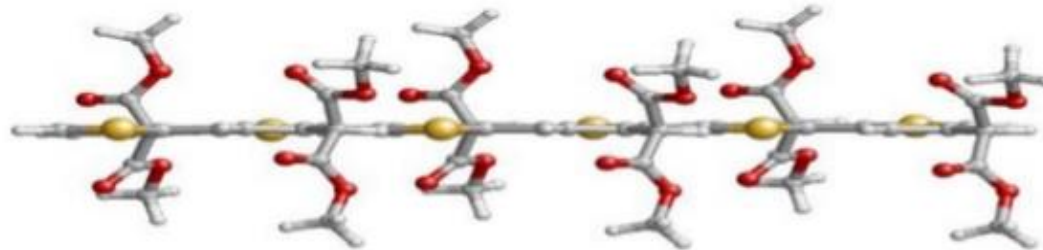
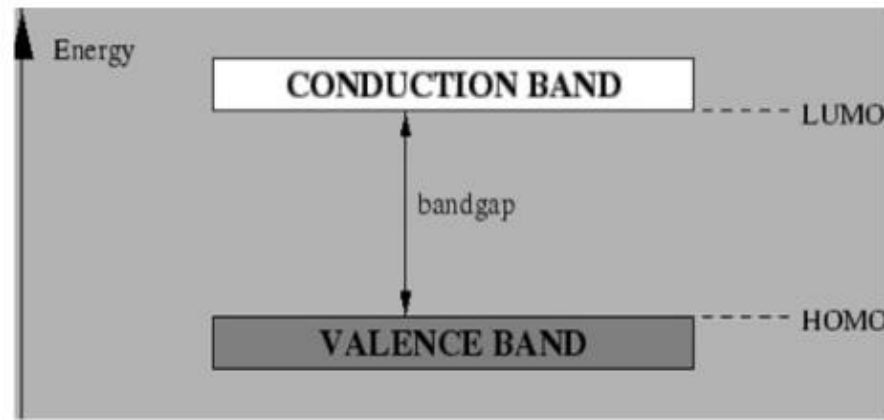
As a class, polymers are known for **electrical insulation** property because the organic molecules of which they are composed have no **free electrons** to carry current ; all the electrons are held firmly by atoms forming the molecules

In the (1950) polymers were made conducting by incorporation of a critical proportion of conducting charge particles such as metallic powder, **metal fibers, carbon black etc.** these conducting materials provided some electronic conduction.

Recent research shown that organic polymers can be made intrinsically conductive by structural modifications during polymerization or by doping which make them highly suitable for use in semiconductor devices

Mechanism of Conduction

It is generally agreed that the mechanism of conductivity in these polymers is based on the motion of charged defects within the conjugated framework. The charge carriers, either positive p-type or negative n-type, are the products of oxidizing or reducing the polymer respectively.



How they conduct ?

Conducting polymers have a continuous chain of SP^2 hybridized carbon centers. It is their π bonds due to which they conduct.

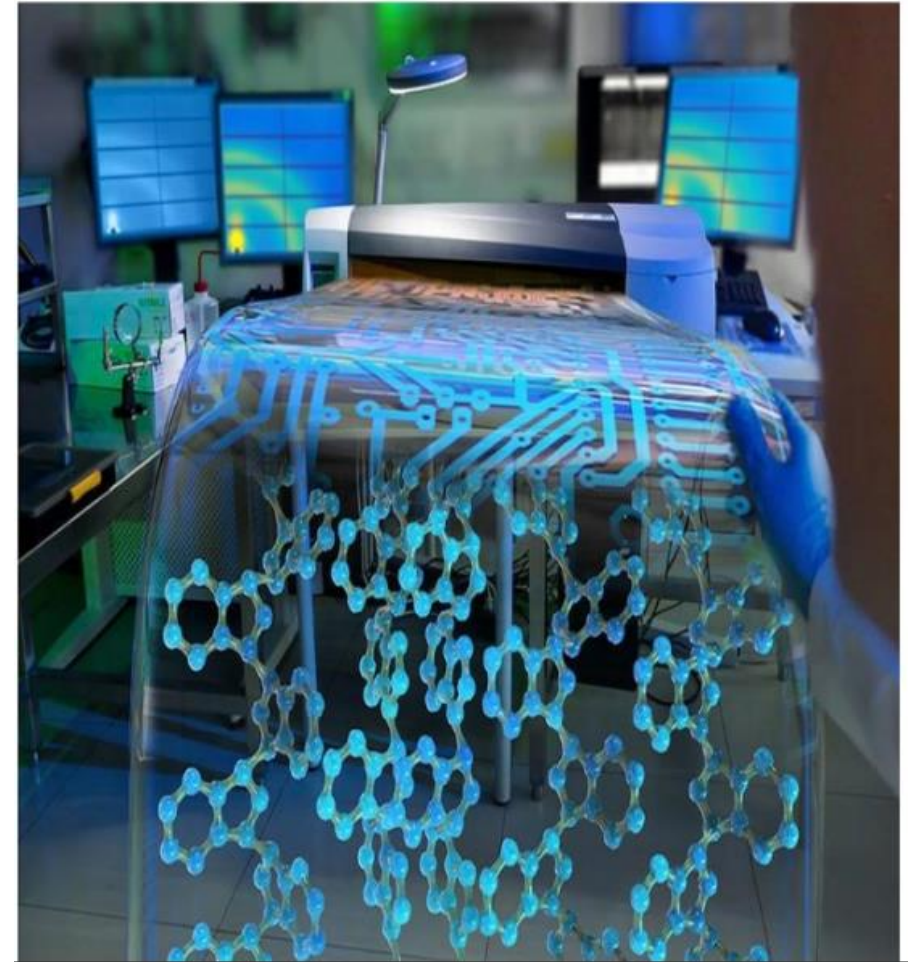
Two conditions

- (i) Presence of conjugated double bonds
- (ii) Molecule must be distributed using a dopant

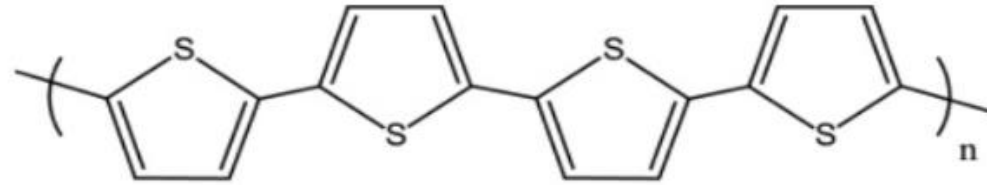
Oxidative doping (halogens) P-type

Semiconducting polymers

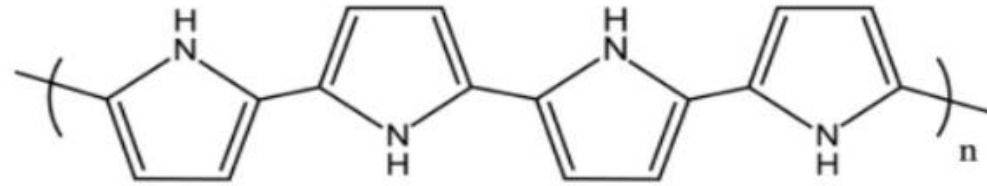
Reductive doping (alkali metals) N-type



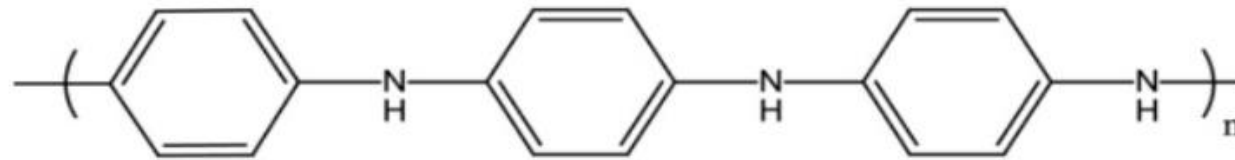
Examples of conductive polymers



polythiophene



polypyrrole



polyaniline

1. Polypyrroles

A type of organic polymer formed by polymerization of pyrrole

Pyrrole is a heterocyclic aromatic organic compound, a five-membered ring with the formula C_4H_4NH

Polypyrrole films are prepared with p-type conductivities of up to 100 S cm^{-1} by electrolytic oxidation of pyrrole in acetonitrile solution using either tetraethylammonium tetrafluoroborate or silver perchlorate

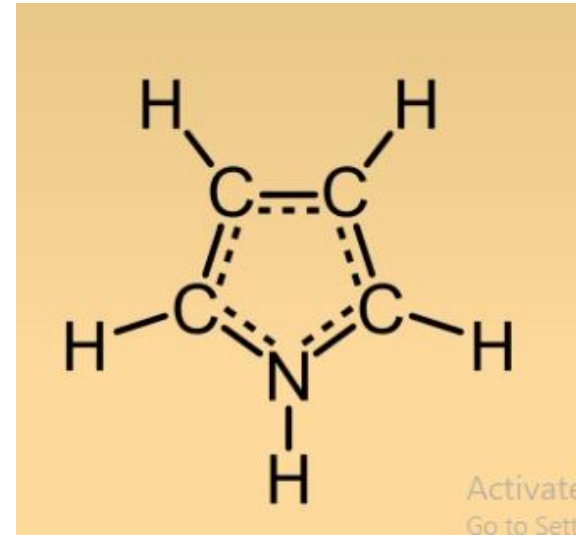


Fig. 3 Repeat unit of polypyrrole

Common fabrication method

- Common Preparation Method

Most commonly Ppy is prepared by oxidation of pyrrole, which can be achieved using ferric chloride in methanol:

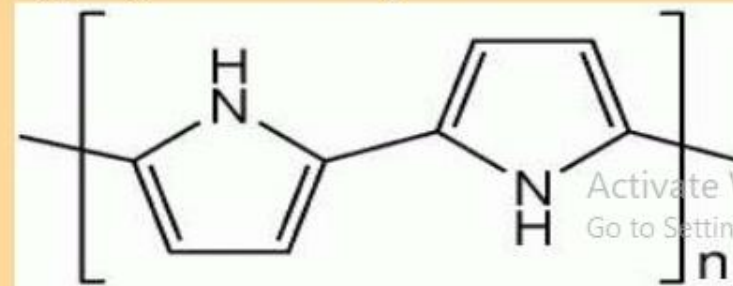


- Polymerization of Conductive form

Conductive forms of PPy are prepared by oxidation ("p-doping") of the polymer:



1. "Polypyrrole: a conducting polymer; its synthesis, properties and applications" Russ. Chem. Rev. 1997, vol. 66, p.443.



- Charge carriers in polypyrrole results when the neutral, non-conductivity polymer is oxidized for conducting polymer, the conductivity is both temperature and frequency dependent
- Solubility of polypyrrole is limited due to its rigid structure and cross linking
- Polypyrrole can be deposited directly onto a substrate surface, by placing the object in a solution containing pyrrole and oxidant
- The films are flexible, non fibrous and shiny and are black in colour.
- The interesting point is that they are stable in air and can be heated upto 250 °C with little effect on charge transport process
- They possess good mechanical properties, low density and resistant to moisture

Applications

1. Good applications in conducting and semiconducting circuits for electronic devices
2. Radar absorbers or chemical sensors
3. Potential vehicle for drug delivery
4. Artificial muscles

Disadvantages

1. Relatively high cost
2. Difficult in processing
3. Lack of mechanical stability after doping
4. Difficult to fabricate

